

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT3 — CASE STUDY

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO2 Statement:	<i>Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them. (BL5)</i>		
Student Name:	RAGUL.M	Reg. No.:	192524066
Section / Batch:	SLOT A	Date:	7/08/2026

Code of Conduct

I RAGUL.M {192524066} that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application	20%	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with well-reasoned conclusions.	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.
Organization	15%	The response is clearly structured, addressing every sub-part (a–d) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CASE STUDY 4 — Sports Team Performance

a) What type of chart shows win/loss composition per team most clearly?

A stacked bar chart is the best choice for showing the win/loss composition of each team.

- Each bar represents one team.
- The bar is divided into Wins and Losses.
- The length of each section shows the number of wins or losses.
- It allows easy comparison of both total matches and team performance.
- A common baseline makes comparison between teams clear.

Answer: Stacked bar chart is suitable because it clearly displays the composition of wins and losses for every team.

b) Which chart type and coordinate system is best for comparing the top team's five attributes at once?

A radar chart (spider chart) with a polar coordinate system is suitable.

- The five attributes are represented along five radial axes.
- Each attribute extends outward from the center.
- The top team's values are plotted and connected to form a polygon.
- The polar coordinate system makes it easy to compare multiple attributes simultaneously.
- It provides a compact visual profile of the team's strengths and weaknesses.

Answer: A radar chart using a polar coordinate system is best for comparing five attributes of one team.

c) Dashboard combining team ranking and top-team attribute profile

The dashboard can contain:

1. Horizontal bar chart – displays all 10 teams according to their points/ranking.
2. Radar chart – displays the five attributes of the highest-ranked team.
3. KPI card – highlights the top team's total points and rank.
4. Annotation – identifies the strongest attribute of the top team.

Narrative takeaway:

“The top-ranked team leads the competition through consistently strong performance across its five key attributes.”

The two charts should use a consistent theme and place the ranking chart on the left and the radar chart on the right.

d) Would the radar chart work well for all 10 teams?

No. A radar chart would become difficult to interpret when all 10 teams are displayed together.

- Too many overlapping polygons would create visual clutter.
- Similar values would be difficult to distinguish.
- The chart would make comparison between individual teams difficult.
- It would reduce readability and increase cognitive load.

A better alternative is a heatmap.

In a heatmap:

- Rows represent the 10 teams.
- Columns represent the five attributes.
- Cell intensity represents the attribute value.
- All teams can be compared quickly across all five attributes.

Answer: Radar chart is effective for one team but not for 10 teams. A heatmap is a better alternative for comparing all teams.



Tableau Public - Book2

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help



Dashboard

Layout <

Default

Phone

Device Preview

Size

min 420x560 - max 650x860

Sheets

Sheet 1

Sheet 2

Sheet 3

Objects

Horizontal Container

Vertical Container

Text

Extension

Pulse Metric

Image

Tiled

Floating

Show dashboard title

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Measure Names

Losses

Wins

Points

28 48

192524066

Team

Team J

Team I

Team H

Team G

0 10 20 30 40 50

Points

192524066

Team

Team A

Team B

Team C

Team D

Team E

Team F

Team G

Team H

Team I

Team J

0 5 10 15 20 25

Value

CASE STUDY 9 — Supplier & Production Analysis

a) What axis scale is appropriate for the district spending data, and why?

A linear axis scale is appropriate for district spending data when the values are ordinary quantitative amounts with no extreme exponential growth.

- Spending is numerical and quantitative.
- A linear scale represents equal differences equally.
- It makes direct comparison between districts easy.
- The axis should start at zero when possible to avoid misleading comparisons.

Answer: Use a linear scale because district spending is quantitative data where equal numerical differences should appear as equal visual distances.

b) Which chart correctly represents ordinal service-satisfaction categories while preserving their order?

A sorted bar chart is the most appropriate.

The categories should remain in their natural ordinal order:

Low → Medium → High

- Bar length represents the number or percentage of responses.
- The natural order is preserved.
- It avoids treating ordinal categories as unrelated nominal categories.
- It is easier to compare than a pie chart.

Answer: A bar chart with the categories ordered Low, Medium, High correctly represents ordinal satisfaction data.

c) Chart and color scale for year-over-year spending changes

A diverging bar chart is suitable for year-over-year spending changes.

- Positive values extend above the zero baseline.
- Negative values extend below the zero baseline.
- Zero represents no change.
- Use a diverging color scale with one color family for increases and another for decreases.
- The intensity can represent the magnitude of the change.

For example:

Increase → positive side

No change → neutral midpoint

Decrease → negative side

Answer: Use a diverging bar chart with a diverging color scale because the data contains two meaningful directions: increases and decreases.

d) Combined dashboard with actionable narrative

A one-page dashboard can combine:

1. District Spending Chart

- Horizontal bar chart showing spending by district.
- Allows quick identification of districts with the highest and lowest spending.

2. Year-over-Year Change Chart

- Diverging bar chart showing increases and decreases.
- A zero reference line separates increases from decreases.

Dashboard coordination:

- Use the same district names in both views.
- Maintain consistent ordering.
- Use a common visual theme.
- Add an annotation to the district with the largest spending increase or decrease.



Tableau Public - Book1

File
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Analysis
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Dashboard
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Default
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Device Preview

Size
min 420x560 - max 650x860

Sheets
Sheet 1
Sheet 2
Sheet 3

Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Matrix

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YOY spending cngange
-16
24

Satisfaction

Satisfaction Level	Satisfaction Count
Low	320
Medium	600
High	880

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District

District	YOY spending cngange
District A	24
District B	13
District C	4
District D	-5

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Distr..

District	Current Spending
District F	60
District E	75
District D	86
District C	90